

UNIVERSITY OF OTAGO EXAMINATIONS

STATISTICS
STAT115
Introduction to Biostatistics
Semester 2

(TIME ALLOWED: 3 HOURS)

This examination paper comprises 31 pages

Candidates should answer questions as follows:

This is a multiple-choice examination with 92 questions.

Select ONE answer only from the 5 options provided for each question.

Record your answers on the machine-readable sheet provided.

All questions should be attempted.

Use the blank reverse side of the examination paper pages for rough work.

The following material is provided:

A detachable appendix giving a summary of formulae.

Use of calculators:

Any model of calculator can be used for this exam. Provided that it is battery powered, truly portable, silent and free of communication capabilities.

Candidates are permitted copies of: NIL

Other Instructions:

Hand in the entire paper and appendix attached to the MCQ answer sheet and green attendance receipt.

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Information for Questions 1 to 3

A sample is carried out by an intermediate school-aged class to estimate the average weight of students of that age group. The weights of the five students (in kg) in the sample were:

36 49.1 41.5 40 46.4

1. What is the mean of this sample closest to?
A) 46.9kg B) 42.6kg C) 51.1kg D) 44.3kg E) 42.9kg
2. What is the standard deviation for these data closest to?
A) 27.1kg B) 6.4kg C) 21.6kg D) 4.7kg E) 5.2kg
3. Choose the answer that best describes the type of data in this question.
A) Nominal. C) Continuous. E) Numerical binary.
B) Discrete numerical. D) Categorical ordinal.

Information for Questions 4 to 5

Some researchers are interested in the average number of minutes per week that New Zealand high school students spend conducting moderate to high-intensity cardiovascular exercise.

A survey they conducted involving 764 students found that, on average, the students spent 65 minutes involved in moderate to high-intensity cardiovascular exercise per week.

4. The quantity '65 minutes' is an example of a
A) Statistic and an observed value of a random variable
B) Statistic but not an observed value of a random variable
C) Parameter and an observed value of a random variable
D) Parameter but not an observed value of a random variable.
E) Statistic, parameter, and observed value of a random variable
 5. What is the study population of interest and what is the parameter the researchers are interested in estimating in this population in this survey?
A) The study population is the 764 New Zealand high school students surveyed, and the parameter of interest is the total number of minutes per week these students spent in moderate to high-intensity cardiovascular exercise in the survey.
B) The study population is the 764 New Zealand high school students surveyed, and the parameter of interest is the average number of minutes per week these students spent in moderate to high-intensity cardiovascular exercise in the survey.
C) The study population is New Zealand high school students, and the parameter of interest is the average number of minutes per week that they spend conducting moderate to high-intensity cardiovascular exercise.
D) The study population is New Zealand high school students, and the parameter of interest is the total number of minutes per week that they spend conducting moderate to high-intensity cardiovascular exercise.
E) None of the above.
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Information for Questions 6 to 9

A prospective study was conducted examining the occurrence rates of osteoporotic fractures among elderly men and women in an Australian town between 1989 and 2000 (including the years 1989 and 2000). Rates of fractures were obtained by examining radiology reports from the town's radiology practices.

Some of the data are summarised in the table below:

Period	Sex	Number of hip fractures	Population
1989-2000	Female	175	1904
1989-2000	Male	54	1373

6. Which of the following is a correct statement about the frequency of hip fractures among women.
 - A) The prevalence of hip fractures among women in the study is approximately 765.9 fractures per 100,000 people per year.
 - B) The incidence rate of hip fractures among women in the study is approximately 765.9 fractures per 100,000 people per year.
 - C) The incidence rate of hip fractures among women in the Australian town studied is approximately 765.9 fractures per 100,000 people per year.
 - D) The prevalence of hip fractures among women in the study is approximately 9191.2 fractures per 100,000 people per year.
 - E) The incidence rate of hip fractures among women in the study is approximately 9191.2 fractures per 100,000 people per year.

7. The relative risk of women developing hip fractures compared to men developing hip fractures in the study is closest to
 - A) 0.43
 - B) 2.34
 - C) 0.43 per 100,000 people per year
 - D) 2.34 per 100,000 people per year
 - E) 1

8. If men are less likely to get potential hip fractures diagnosed, which of the below statements is true regarding the effect this would have on the estimate of relative risk?
 - A) This would bias the estimate of relative risk calculated in the study from the true relative risk, but not in any particular direction.
 - B) This would bias the estimate of relative risk calculated in the study to be more likely to be lower than the true relative risk.
 - C) This will not bias the estimate of the relative risk, because sex is controlled for in the analysis.
 - D) This would bias the estimate of relative risk calculated in the study to be more likely to be greater than the true relative risk.
 - E) This would not bias the estimate of relative risk, because the rates of hip fractures are taken from official radiology records.

9. In the study data, the researchers also found that the relative risks of hip fracture between women and men differed based on age: They found that the relative risk for the 60-69 age group was 4.5, and the relative risk for the 80+ age group was 1.9. Which of the below is an accurate interpretation of what this means.
- A) In the study, the rate of hip fracture per person per year was 4.5 times higher in women than men in the 60-69 age group, and 1.9 times higher in women than men in the 80+ age group.
 - B) In the population, the rate of hip fracture per person per year was 4.5 times higher in women than men in the 60-69 age group, and 1.9 times higher in women than men in the 80+ age group.
 - C) Women incur 2.37 times more hip fractures per person per year in the 60-69 age group than in the 80+ age group.
 - D) Men incur 2.37 times more hip fractures per person per year in the 80+ group than in the 60-69 age group.
 - E) None of the above.
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Information for Questions 10 to 11

The MAADI cup is the annual New Zealand secondary schools' rowing championship.

Let X be the number of students who attend this event each year from the North Island, and Y be the number who attend the event from the South Island.

Thus the total number of competitors who attend the event from both islands is $W = X + Y$

The mean of X , μ_X , is 1750, and the standard deviation of X , σ_X , is 100.

The mean of Y , μ_Y , is 750, and the standard deviation of Y , σ_Y , is 40.

You may assume X and Y are independent.

10. What is μ_W , the mean number of students who attend the event?
- A) 2600 B) 4250 C) 2150 D) 2500 E) 2400
11. What is σ_W , the standard deviation of the number of students who attend the event?
- A) 140.00 B) 11.83 C) 107.70 D) 16.73 E) 11600.00
-

Information for Questions 12 to 14

The National Alliance on Mental Illness were interested in exploring the overlap between anxiety related disorders and ADHD. The results from 623 randomly selected American adults, are summarised below:

		ADHD		
		Yes (B)	No ($\bar{B}$)	Total
Anxiety Disorder	Yes (A)	32	113	145
	No ($\bar{A}$)	50	428	478
Total		82	541	623

In the questions below, let A be the event "has anxiety", and B be the event "has ADHD".

12. Calculate the probability that a randomly selected study participant had ADHD, i.e. $\Pr(B)$.
 A) 0.0803 B) 0.1814 C) 0.1316 D) 0.2327 E) 0.6098
13. Calculate the probability that a randomly selected study participant had ADHD but not an anxiety related disorder, i.e. $\Pr(\bar{A} \cap B)$.
 A) 0.0803 B) 0.1814 C) 0.0514 D) 0.2327 E) 0.6098
14. Calculate the probability that a randomly selected study participant with ADHD did not have an anxiety related disorder, i.e. $\Pr(\bar{A} | B)$.
 A) 0.0803 B) 0.1814 C) 0.0514 D) 0.3902 E) 0.6098

Information for Questions 15 to 20

Digital mammography is believed to be an effective tool for detecting breast cancer. Digital mammography will return a positive result for 97% of women with breast cancer and 35% of women without it. 20% of women who undergo digital mammography tests for breast cancer have breast cancer.

Let B be the event that a woman has breast cancer, and T be the event that a woman tests positive for breast cancer.

15. What is the sensitivity, $\Pr(T|B)$, of this test?
 A) 0.2 B) 0.97 C) 0.03 D) 0.35 E) 0.65
16. What is the specificity, $\Pr(\bar{T}|\bar{B})$, of this test?
 A) 0.65 B) 0.35 C) 0.03 D) 0.97 E) 0.2
17. What is the probability that a woman taking the test has both breast cancer and a positive test, i.e. what is $\Pr(B \cap T)$?
 A) 0.07 B) 0.194 C) 0.13 D) 0.52 E) 0.776
18. What is the probability that a woman taking the test does not have breast cancer and has a positive test, i.e. what is $\Pr(\bar{B} \cap T)$?
 A) 0.194 B) 0.13 C) 0.28 D) 0.52 E) 0.776

19. What is the probability that a woman taking the test has a positive test i.e. what is $\Pr(T)$?
- A) 0.47 B) 0.97 C) 0.35 D) 0.66 E) 0.26
20. What is the positive predictive value, $\Pr(B|T)$, of this test?
- A) 0.59 B) 0.01 C) 0.41 D) 1.10 E) 0.97
-

Information for Questions 21 to 23

A district with several mountains is renowned for being a high-risk zone for avalanches. A geologist is trying to predict the number of mountains in the district that will have an avalanche in a given year.

She decides to use a binomial model to estimate the number of mountains in the district that will have one or more avalanches in a given year.

21. Letting X be the number of mountains in the district that have an avalanche in a given year, select the conditions that must apply for X to be a binomial random variable.
- A) Fixed number of mountains in the district; the probability each mountain has an avalanche differs across all mountains; the probability that each mountain has an avalanche depends on whether some of the other mountains have had an avalanche; exactly two possible outcomes for each mountain.
- B) Fixed number of mountains in the district; the probability each mountain has an avalanche is the same across all mountains; the probability that each mountain has an avalanche does not depend on whether any of the other mountains have had an avalanche; exactly two possible outcomes for each mountain.
- C) Two mountains in the district; the probability each mountain has an avalanche is between 0 and 1; the probability that each mountain has an avalanche does not depend on whether any of the other mountains have had an avalanche; multiple possible outcomes for each mountain.
- D) Two mountains in the district; the probability each mountain has an avalanche is the same across all mountains; the probability that each mountain has an avalanche depends on whether some of the other mountains have had an avalanche; multiple possible outcomes for each mountain.
- E) Fixed number of mountains in the district; the probability each mountain has an avalanche is between 0 and 1; the probability that each mountain has an avalanche does not depend on whether any of the other mountains have had an avalanche; One possible outcome for each mountain.

Suppose the district has 5 mountains, and the geologist assumes the probability each mountain has an avalanche is $\pi = 0.1$. The table below displays the probability distribution for the number of mountains in the district that have an avalanche with $n = 5$ and $\pi = 0.1$:

x_i	0	1	2	3	4	5
$\Pr(X = x_i)$	0.59049	0.32805	0.0729	0.0081	0.00045	0.00001

22. The probability that exactly 3 of the mountains have an avalanche is:

- A) 0.32805 C) 0.0081 E) 0.59049
 B) 0.0729 D) 0.00045

23. The probability that more than 3 of the mountains have an avalanche is:

- A) 0.00856 B) 0.00001 C) 0.0081 D) 0.00046 E) 0.99954

Information for Questions 24 to 27

The systolic blood pressures of adults are normally distributed with a mean of 127.5 mmHg and a standard deviation of 19.6 mmHg.

24. Suppose you were asked to calculate the probability an individual has a blood pressure above 150 mmHg. Which of the statements below could be made without using any software.

- A) Because the sample size is unknown, we cannot say, because the sample size is needed to calculate the standard error.
 B) We cannot know the probability with the information we have, because we need a confidence interval for the systolic blood pressure to know the probability an individual has a systolic blood pressure within any specific range.
 C) The probability will be 0, because, on a normally distributed curve with mean 127.5 mmHg and standard deviation 19.6 mmHg, it is impossible for a person to have a systolic blood pressure as far away from the mean as 150mmHg.
 D) The probability will be below 0.5 but above 0.
 E) The probability will be above 0.5.

25. Select the correct R command to find the probability a randomly selected individual has a systolic blood pressure above 137.5mmHg.

- A) `1 - pnorm(137.5, mean = 127.5, sd = 19.6)`
 B) `pnorm(137.5, mean = 127.5, sd = 19.6)`
 C) `1 - pnorm(137.5, mean = 127.5, sd = 19.6/√n)`
 D) `pnorm(137.5, mean = 127.5, sd = 19.6/√n)`
 E) `pnorm(137.5)`

26. Select the correct R command to find the probability a randomly selected person has a systolic blood pressure between 119.5 and 134.5mmHg.
- A) `pnorm(119.5, 127.5, 19.6) – pnorm(134.5, 127.5, 19.6)`
 - B) `pnorm(134.5, 127.5, 19.6) – pnorm(119.5, 127.5, 19.6)`
 - C) `1 – pnorm(134.5, 127.5, 19.6) – pnorm(119.5, 127.5, 19.6)`
 - D) `pnorm(134.5, 127.5, 19.6/√n) – pnorm(119.5, 127.5, 19.6√n)`
 - E) `pnorm(134.5) – pnorm(119.5)`
27. Calculate how many standard deviations from the mean an individual is if they have a systolic blood pressure of 140.5mmHg, i.e. find the z -value that corresponds to a systolic blood pressure of 140.5mmHg.
- A) -0.66
 - B) 2.00
 - C) 1
 - D) 0.66
 - E) 13

Information for Questions 28 to 29

Suppose we took a sample of 195 New Zealand adult males and measure their BMI, and the mean BMI in the sample was 28kg/m^2 , while the standard deviation was 5.9kg/m^2 . If we constructed the two intervals below using an α value of $\alpha = 0.05$, select the appropriate interpretations of the intervals.

You may assume that the BMI of New Zealand adult males is normally distributed.

$$\text{Interval A : } 28 \pm t_{\left(1-\frac{\alpha}{2}, 194\right)} \times \frac{5.9}{\sqrt{195}}$$

$$\text{Interval B : } 28 \pm z_{\left(1-\frac{\alpha}{2}\right)} \times 5.9$$

28. Which of the following does Interval A represent?
- A) A 95% confidence interval. We estimate the BMI for the central 95% of all NZ adult males is in this range.
 - B) A 95% confidence interval. We can be 95% confident that the mean BMI across all NZ adult males is in this range.
 - C) A 95% reference range. We can be 95% confident that the mean BMI across all NZ adult males is in this range.
 - D) A 95% reference range. We estimate the BMI for the central 95% of all NZ adult males is in this range.
 - E) The interval has no discernable meaning.
29. Which of the following does Interval B represent?
- A) A 95% confidence interval. We estimate the BMI for the central 95% of all NZ adult males is in this range.
 - B) A 95% confidence interval. We can be 95% confident that the mean BMI across all NZ adult males is in this range.
 - C) A 95% reference range. We can be 95% confident that the mean BMI across all NZ adult males is in this range.
 - D) A 95% reference range. We estimate the BMI for the central 95% of all NZ adult males is in this range.
 - E) The interval has no discernable meaning.

Information for Questions 30 to 33

A study examined whether a sauna bathing protocol could improve results in long distance runners. One such indicator of results in long distance runners is their resting heart rate, with a lower resting heart rate indicating better fitness.

The study featured 13 female runners who didn't currently use saunas, and was conducted over 7 weeks. They tested their resting heart rates before and after the study.

The mean resting heart rate before the study was 56.5bpm, while the mean immediately following the study was 52.9bpm. The standard deviation of the differences in the sample were 5.9bpm.

You may assume the differences in resting heart rates are normally distributed.

30. Find the mean of the paired differences in resting heart rate from before the study to after the study (hint: a mean resting heart rate that is lower after than before will result in a negative mean difference, while one that is higher than it was before will result in a positive mean difference).
- A) 5.9 bpm B) 3.6 bpm C) -3.6 bpm D) 0.61 bpm E) -0.61 bpm
31. What is the estimated standard error of the differences calculated from this sample?
- A) 5.9 bpm C) 10.05 bpm E) 1.64 bpm
B) 9.65 bpm D) 0.45 bpm
32. State the correct R command for obtaining the appropriate multiplier for calculating a 95 confidence interval for the mean difference in resting heart rates before and after the sauna protocol.
- A) **qt(0.975, 12)** C) **qt(0.95, 12)** E) **qt(0.975, 13)**
B) **pt(0.975, 12)** D) **qnorm(0.975)**

33. The confidence interval for the difference in means is $(-7.17, -0.03)$. Which of the below is the correct interpretation of this confidence interval.
- A) Across our study group, we can be 95% confident that the mean change in heart rate between before and after the sauna protocol lied within the limits above. Since the interval contains 0, the data provide insufficient evidence that there was a change in mean resting heart rate among study participants who undertook the sauna protocol.
 - B) Across our study group, we can be 95% confident that the mean change in heart rate between before and after the sauna protocol lied within the limits above. Since the interval does not contain 0 and is entirely negative, the data provide evidence that there was a decrease in mean resting heart rate among study participants who undertook the sauna protocol.
 - C) Across female runners, we can be 95% confident that the mean difference in resting heart rate between before and after the sauna protocol lies in the limits above. Since 0 is in the interval, the data provide insufficient evidence that the mean resting heart rate changes among female runners who undergo the sauna protocol.
 - D) Across female runners, we can be 95% confident that the mean difference in resting heart rate between before and after the sauna protocol lies in the limits above. Since 0 is not in the interval, the data provide evidence that the mean resting heart rate changes among female runners who undergo the sauna protocol. Since the interval is entirely negative, the data provide evidence that the mean resting heart rate among female runners who undergo the sauna protocol decreases.
 - E) Across female runners, we can be 95% confident that the mean difference in resting heart rate between before and after the sauna protocol lies in the limits above. Since 0 is not in the interval, the data provide evidence that the mean resting heart rate changes among female runners who undergo the sauna protocol. Since the interval is entirely negative, the data provide evidence that the mean resting heart rate among female runners who undergo the sauna protocol increases.

Information for Questions 34 to 39

Trust in mainstream media around the western world is suspected to be declining rapidly. In a survey involving 99 New Zealand adults examining this, sociologists found that mainstream media was the main news source for 49 of them.

34. What is the estimate of the proportion of New Zealand adults whose main news source is mainstream media, based on this sample?
- A) 0.495 B) 49.5 C) 49 D) 99 E) 0.505
35. What is the estimated standard error (SE) for the proportion of New Zealand adults whose main news source is mainstream media, based on this sample?
- A) 24.75 B) 0.050 C) 0.250 D) 0.025 E) 0.003

36. Let p be the proportion whose main news source is mainstream media in the sample, SE be the estimated standard error calculated in the previous question, and `pnorm`, `qnorm`, `pt`, and `qt` refer to the R commands of those names. Which of the below is the correct formula for calculating the 95% confidence interval for the proportion of New Zealand adults for whom mainstream media is their main news source based on this sample.
- A) $p \pm \text{pnorm}(0.95) \times SE$ D) $p \pm \text{pt}(0.95, 98) \times SE$
B) $p \pm \text{qnorm}(0.975) \times SE$ E) $p \pm \text{qt}(0.975, 98) \times SE$
C) $p \pm \text{qnorm}(0.95) \times SE$
37. The 95% confidence interval for the proportion of New Zealand adults whose main news source is mainstream media is $(0.40, 0.59)$. Give the correct interpretation of what this means below.
- A) Across 95% of New Zealand adults, the proportion whose main news source is mainstream media is between 0.40 and 0.59.
B) We are 95% confident that the proportion of New Zealand adults in our sample whose main news source is mainstream media is between 0.40 and 0.59.
C) We are 95% confident that the proportion of New Zealand adults whose main news source is mainstream media is between 0.40 and 0.59.
D) We are 95% confident that the proportion of adults in the world whose main news source is mainstream media is between 0.40 and 0.59.
E) We are 95% confident that the proportion of adults whose main news source is mainstream media is above 0.40.
38. Which assumption is required for the confidence interval calculated above to be valid.
- A) The random variable for whether each person gets their news from mainstream media follows a normal distribution.
B) The random variable for whether each person gets their news from mainstream media follows a t -distribution.
C) The sample size is large enough, and the proportion who get their news from mainstream media is close enough to 0.
D) The sample size is large enough, and the proportion who get their news from mainstream media is close enough to 1.
E) The sample size is large enough, and the proportion who get their news from mainstream media is close enough to 0.5.

39. Suppose that people who got their news from mainstream media had a higher response rate than those who didn't. What impact would this likely have on the estimated proportion?
- A) This is an example of selection bias. The proportion surveyed who got their news from mainstream media would be biased towards being an underestimate of the population proportion.
 - B) This is an example of selection bias. The proportion surveyed who got their news from mainstream media would be biased towards being an overestimate of the population proportion.
 - C) This is an example of selection bias. The proportion surveyed who got their news from mainstream media would be biased, but not in any particular direction.
 - D) This is an example of information bias. The proportion surveyed who got their news from mainstream media would be biased towards being an underestimate of the population proportion.
 - E) This is an example of information bias. The proportion surveyed who got their news from mainstream media would be biased towards being an overestimate of the population proportion.
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Information for Questions 40 to 44

Quitline (a New Zealand agency that helps people quit smoking) are interested in quantifying the change in daily smoking rates among New Zealand adults (those aged 18 and over) between 2010 and 2020.

A survey involving $n_1 = 970$ adults in 2010 found that $x_1 = 180$ of them smoked daily, while another independent survey in 2020 involving $n_2 = 990$ adults found that $x_2 = 110$ of them smoked daily.

40. Which of the below best describes the type of study conducted here?
- A) Analytic randomised control trial.
 - B) Analytic cohort study.
 - C) Descriptive study.
 - D) Analytic case control study.
 - E) Descriptive cohort study.
41. Find the change in the proportion of study participants who smoked between the 2010 and the 2020 surveys (i.e. letting p_1 be the proportion who smoked in 2010 and p_2 be the proportion who smoked in 2020, find $p_2 - p_1$).
- A) -0.074
 - B) -7.45
 - C) -0.084
 - D) -6.45
 - E) -0.064

42. How would you calculate the standard error (SE) for the difference between the proportions of New Zealand adults who smoked in 2010 and 2020?

- A) $\sqrt{\frac{0.1856^2}{990} + \frac{0.1111^2}{970}}$
- B) $\sqrt{\frac{180}{970} + \frac{110}{990}}$
- C) $\sqrt{\frac{0.1856(1 - 0.1856)}{990} + \frac{0.1111(1 - 0.1111)}{970}}$
- D) $\sqrt{\frac{0.1856(1 - 0.1856)}{970} + \frac{0.1111(1 - 0.1111)}{990}}$
- E) $\sqrt{\frac{180(1 - 180)}{970} + \frac{110(1 - 110)}{990}}$

43. State the correct formula for calculating a 99% confidence interval for the change in the proportion of New Zealand adults who smoked between the 2010 and 2020 surveys.

- A) $(p_2 - p_1) \pm t_{(0.99, n-1)} \times SE$
- B) $(p_2 - p_1) \pm t_{(0.995, n-1)} \times SE$
- C) $(p_2 - p_1) \pm z_{(0.99)} \times SE$
- D) $(p_2 - p_1) \pm z_{(0.995)} \times SE$
- E) $(p_2 - p_1) \pm z_{(0.995)} \times SE^2$

44. The 99% confidence interval for the difference between the proportion of New Zealand adults who smoked in 2020 and 2010 from this sample is $(-0.116, -0.033)$. Give a valid interpretation of this from the options below.

- A) We are 99% confident that the difference between the proportion of New Zealand adults who smoked in 2020 and 2010 lies within the limits $(-0.116, -0.033)$. Since this interval does not contain 0 and is entirely negative, this provides evidence that the proportion of New Zealand adults who smoke decreased between 2010 and 2020.
- B) We are 99% confident that the difference between the proportion of New Zealand adults who smoked in 2020 and 2010 lies within the limits $(-0.116, -0.033)$. Since this interval does not contain 0 and is entirely negative, this provides evidence that the proportion of New Zealand adults who smoke increased between 2010 and 2020.
- C) We are 99% confident that the difference between the proportion of New Zealand adults who smoked in 2020 and 2010 lies within the limits $(-0.116, -0.033)$. Since this interval is close to 0, this does not provide sufficient evidence that the proportion of New Zealand adults who smoke changed between 2010 and 2020.
- D) We are 99% confident that difference between the proportion of New Zealand adults who smoked in 2020 and 2010 lies within the limits $(-0.116, -0.033)$. Since this interval contains 0, this does not provide sufficient evidence that the proportion of New Zealand adults who smoke changed between 2010 and 2020.
- E) We are 99% confident that the proportion of New Zealand adults who smoked in 2020 lies within the limits $(-0.116, -0.033)$. Since this interval does not contain 0 and is entirely negative, this provides evidence that the proportion of New Zealand adults who smoke was negative in 2020.

Information for Questions 45 to 52

Mercury is a type of heavy metal that occurs naturally in the environment. Humans consume some in their diet by consuming fish, but humans can also be exposed to it through industrial sources. Levels of mercury higher than 50 ng/mL of blood are deemed to be dangerous in humans.

It is commonly believed that consuming an algae called chlorella can detox mercury from the body. A group of food scientists are interested in testing whether chlorella does indeed detox mercury from the body. They recruit a group of 193 individuals who have recently been found to have dangerous levels of mercury (above 50 ng/mL of blood). They assign $n_C = 91$ participants to a control group that take a placebo pill, and $n_I = 102$ to an intervention group that consume a 3g chlorella pill once daily.

The reduction in mercury levels over the study period is recorded for each participant. Sample summaries are shown in the table below.

	Intervention	Control
Sample size	102	91
Sample mean	9.6	8.3
Sample variance	24	12.5

45. What type of study did the food scientists conduct?
- Observational analytic case-control.
 - Observational analytic cohort study.
 - Experimental analytic randomised control trial.
 - Observational descriptive randomised control trial.
 - Experimental descriptive cohort study.
46. Letting μ_I and μ_C denote the population mean mercury level reductions in those who follow the intervention group protocol and those who follow the control group protocol, and $\bar{x}_I$ and $\bar{x}_C$ be the observed sample mean mercury level reductions, set up the appropriate null and alternative hypotheses to explore the question the researchers are interested in answering.
- $H_0 : \bar{x}_I = \bar{x}_C; H_A : \bar{x}_I \neq \bar{x}_C$
 - $H_0 : \bar{x}_I - \bar{x}_C = 0; H_A : \bar{x}_I - \bar{x}_C \neq 0$
 - $H_0 : \mu_I = \mu_C; H_A : \mu_I \neq \mu_C$
 - $H_0 : \mu_I \neq \mu_C; H_A : \mu_I = \mu_C$
 - $H_0 : \mu_I - \mu_C \neq 0; H_A : \mu_I - \mu_C = 0$
47. The appropriate standard error to use in a hypothesis test for the difference between the mean mercury level reductions is closest to:
- 4.47
 - 0.61
 - 0.37;
 - 4.90
 - 3.54
48. Select which option the test statistic for this analysis is closest to. For the observed difference, use the ordering $\bar{x}_I - \bar{x}_C$ rather than $\bar{x}_C - \bar{x}_I$ (i.e. the mean reduction in the intervention group minus the mean reduction in the control group).
- 2.13
 - 2.13
 - 4.00
 - 0.29
 - 0.29

- TURN OVER

52. Select the statement regarding the power of this hypothesis test that is definitely correct.
- A) Their study would have had greater power if it had included 400 participants in each of the control and intervention groups, and they had set the significance level at $\alpha = 0.01$.
 - B) Their study would have had lower power if it had included 400 participants in each of the control and intervention groups, and they had set the significance level at $\alpha = 0.01$.
 - C) Their study would have had greater power if it had included 400 participants in each of the control and intervention groups, and they had set the significance level at $\alpha = 0.1$.
 - D) Their study would have had lower power if it had included 400 participants in each of the control and intervention groups, and they had set the significance level at $\alpha = 0.1$.
 - E) Their study would have had greater power if it had included 400 participants in each of the control and intervention groups, but changing the significance level would have no effect on the power of the test.

Information for Questions 53 to 61

In recent years in New Zealand, there has been increased awareness around mental health issues among farmers in New Zealand. A study was carried out to examine the rates of major depressive disorder (MDD) among farmers, and whether they differed from those among non-farmers. A group of 500 farmers and 500 non-farmer workers were selected, and public health records were accessed to assess whether the participants had been diagnosed with MDD at any point in that time. The results are shown in the table below.

		Developed MDD		
		Yes	No	Total
Farmer	Yes	53	447	500
	No	27	473	500
Total		80	920	1000

53. Estimate the odds of developing MDD among farmers in the 10-year study period.
- A) 0.1060 B) 0.1186 C) 0.0540 D) 0.0571 E) 1.9630
54. Estimate the odds of developing MDD among the non-farmers in the 10-year study period.
- A) 0.1060 B) 0.1186 C) 0.0540 D) 0.0571 E) 1.9630
55. Estimate the odds ratio of farmers developing MDD compared to non-farmers.
- A) 2.0771 B) 1.9630 C) 0.4814 D) 0.5094 E) 0.0800
56. The estimated standard error for the log of the odds ratio is closest to:
- A) 0.0603 B) 1.2782 C) -1.4046 D) -2.8091 E) 0.2455
57. Given that the 95% confidence interval for the log of the odds ratio is (0.2499, 1.2121), calculate the 95% confidence interval for the true odds ratio comparing the odds of developing depression in farmers and non-farmers is closest to:
- A) (0.2499, 1.2121) C) (-1.3868, 0.1924) E) (-1.3868, 0.1924)
- B) (1.2839, 3.3606) D) (1.7777, 16.2973)

58. Select the correct option regarding how we would appropriately interpret the confidence interval calculated above. In the answers below, UB refers to the upper bound of the confidence interval, and LB refers to the lower bound of the confidence interval.
- A) If LB is greater than 1, this would constitute evidence that farmers have MDD at rates greater than non-farmers. If LB is less than 1 but UB is greater than 1, the data would constitute insufficient evidence of a difference between the MDD rates among farmers and non-farmers. If UB is less than 1, this constitutes evidence that farmers develop MDD at rates lower than non-farmers.
 - B) If LB is greater than 1, this would constitute evidence that farmers developed MDD at lower rates than non-farmers. If LB is less than 1 but UB is greater than 1, the data would constitute insufficient evidence of a difference between the MDD rates among farmers and non-farmers. If UB is less than 1, this constitutes evidence that farmers develop MDD at rates greater than non-farmers.
 - C) The fact that the central estimate of the odds ratio calculated earlier is above 1 provides equally sufficient evidence that the rates of MDD among farmers were greater than non-farmers as if LB was above 1.
 - D) If LB is greater than 0, this would constitute evidence that farmers developed MDD at rates greater than non-farmers. If LB is less than 0 but UB is greater than 0, the data would constitute insufficient evidence of a difference between the MDD rates among farmers and non-farmers. If UB is less than 0, this constitutes evidence that farmers develop MDD at rates lower than non-farmers.
 - E) If LB is greater than 0, this would constitute evidence that farmers developed MDD at lower rates than non-farmers. If LB is less than 0 but UB is greater than 0, the data would constitute insufficient evidence of a difference between the MDD rates among farmers and non-farmers. If UB is less than 0, this constitutes evidence that farmers develop MDD at rates greater than non-farmers.
59. Suppose farmers were less likely to see a medical professional to have MDD diagnosed when they were displaying symptoms. Select the effect this would have on the odds ratios calculated earlier, and the type of bias this would constitute.
- A) This is an example of information bias. This would bias the odds of MDD among farmers relative to non-farmers seen in the study towards being lower than the true odds of MDD among farmers compared to non-farmers.
 - B) This is an example of information bias. This would bias the odds of MDD among farmers relative to non-farmers seen in the study towards being greater than the true odds of MDD among farmers compared to non-farmers.
 - C) This is an example of information bias. There is no evidence that this would bias the study odds ratio away from the true odds ratio in any particular direction.
 - D) This is an example of selection bias. This would bias the odds of MDD among farmers relative to non-farmers seen in the study towards being lower than the true odds of MDD among farmers compared to non-farmers.
 - E) This is an example of selection bias. This would bias the odds of MDD among farmers relative to non-farmers seen in the study towards being greater than the true odds of MDD among farmers compared to non-farmers.

60. Women are known to have higher MDD rates than men, so researchers considered confounding by sex as well. Select the following statement that may be true.
- A) If women and men have different rates of MDD, then sex will confound the relationship between being a farmer and MDD, regardless of whether there is a relationship between being a farmer and sex.
 - B) If farmers are more likely to be male, and males have lower rates of MDD, then sex may confound the relationship between being a farmer and MDD, making the odds ratio of farmers developing MDD relative to non-farmers appear *greater*. Since the odds ratio of MDD among farmers compared to non-farmers is greater than 1, this would *increase* the association between being a farmer and MDD.
 - C) If farmers are more likely to be male, and males have lower rates of MDD, then sex may confound the relationship between being a farmer and MDD, making the odds ratio of farmers developing MDD relative to non-farmers appear *smaller*. Since the odds ratio of MDD among farmers compared to non-farmers is greater than 1, this would *reduce* the association between being a farmer and MDD.
 - D) If farmers are more likely to be male, and males have lower rates of MDD, then sex may confound the relationship between being a farmer and MDD, making the odds ratio of farmers developing MDD relative to non-farmers appear *greater*. Since the odds ratio of MDD among farmers compared to non-farmers is greater than 1, this would *reduce* the association between being a farmer and MDD..
 - E) If farmers are more likely to be male, and males have lower rates of MDD, then sex may confound the relationship between being a farmer and MDD, making the odds ratio of farmers developing MDD relative to non-farmers appear *smaller*. Since the odds ratio of MDD among farmers compared to non-farmers is greater than 1, this would *increase* the association between being a farmer and MDD.
61. The researchers used logistic regression to estimate the odds ratio for the odds of developing MDD among farmers relative to the odds of developing MDD among non-farmers, adjusting for several variables including age and sex. The resulting estimate was 1.37 and the 95% confidence interval (0.9, 2.78). The best interpretation of this result is:
- A) The adjusted odds ratio decreased to be closer to 1, but since it is still greater than 1, there is still good evidence of an association between being a farmer and MDD, with farmers being more likely to have MDD.
 - B) The adjusted odds ratio decreased to be closer to 0, but since it is still greater than 0, there is still good evidence of an association between being a farmer and MDD, with farmers being more likely to have MDD.
 - C) The adjusted odds ratio increased to be further away from 1. Since 1 is within the confidence interval now however, the data does not sufficient evidence of there being an association between being a farmer and risk of MDD.
 - D) The adjusted odds ratio decreased to be closer to 1. Since 1 is within the confidence interval now however, the data does not provide sufficient evidence of there being an association between being a farmer and risk of MDD.
 - E) The adjusted odds ratio decreased to be closer to 0. But since 0 is outside the confidence interval, the data now provides evidence of there being an association between being a farmer and risk of MDD.
-

Information for Questions 62 to 66

A group of epidemiologists were interested in exploring whether there is an association between the risk of a child developing autism and their father's age at birth. They examined records from an Israeli health database, and observed the following results:

		Father Age in Years			
		20-29	30-39	40+	Total
Diagnosed with Autism	Yes	4	5	11	20
	No	726	523	343	1592
Total		730	528	354	1612

They then conducted a χ^2 -test for independence using this data.

62. Select the appropriate null and alternate hypotheses for the question the epidemiologists are interested in answering.
- A) H_0 : There is an association between whether a child is autistic and their father's age at birth; H_A : There is no association between whether a child is autistic and their father's age at birth.
- B) H_0 : Whether a child is autistic is dependent on their father's age at birth; H_A : Whether a child is autistic is independent of their father's age at birth.
- C) H_0 : There is no association between whether a child is autistic and their father's age at birth; H_A : The older a father is, the greater the chance of their children being autistic.
- D) H_0 : The older a father is when their child is born, the lower the chance of their child being autistic.; H_A : The older a father is when their child is born, the greater the chance of their child being autistic.
- E) H_0 : There is no association between whether a child is autistic and their father's age at birth; H_A : There is an association between whether a child is autistic and their father's age at birth.
63. Select the appropriate expected number of children who developed autism when their fathers were between 30 and 39 at birth, under the null hypothesis.
- A) 6.55 B) 521.45 C) 720.94 D) 9.06 E) 4.39
64. The test statistics follow a χ^2 -distribution with ν degrees of freedom under the null hypothesis, where ν takes which value?
- A) 6 B) 2 C) 1 D) 12 E) 5
65. The χ^2 test statistic for this hypothesis test is 13.30. Letting ν denote the appropriate degrees of freedom, select the appropriate R code to calculate the p -value.
- A) `pchisq(13.30,  $\nu$ )`
- B) `qchisq(13.30,  $\nu$ )`
- C) `1 - pchisq(13.30,  $\nu$ )`
- D) `1 - pchisq(13.30,  $\nu$ , lower.tail=FALSE)`
- E) `2 * (1 - pchisq(13.30,  $\nu$ ))`

66. The p -value calculated from the test statistic is 0.0014. Select the appropriate interpretation to the hypothesis test using $\alpha = 0.05$ as the significance level.
- A) We have evidence that there is an association between a father's age at the birth of their child and whether their child is autistic at the 0.05 significance level.
 - B) We have no evidence that there is an association between a father's age at the birth of their child and whether their child is autistic at the 0.05 significance level.
 - C) We have evidence that father's age at the birth of their child and whether their child is autistic are conditionally independent at the 0.05 significance level.
 - D) There is evidence of a difference between father's age the birth of their child and whether their child is autistic at the 0.05 significance level.
 - E) We have evidence that there is no association between a father's age at the birth of their child and whether their child is autistic at the 0.05 significance level.

Information for Questions 67 to 79

Higher retinol plasma levels are believed to be associated with a lower risk of cancer. A study featuring $n = 100$ New Zealand adults was carried out to explore whether retinol plasma levels (measured in ng/mL) were somewhat dependent on age.

Results from the regression analysis in R are shown below:

```
> summary(lm(retplasma~age))

Call:
lm(formula = retplasma ~ age)

Residuals:
    Min       1Q   Median       3Q      Max
-397.93 -107.56  -23.44  126.86  549.63

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  441.861     66.794   6.615 1.97e-09 ***
age           3.070      1.267   2.424  0.0172 *
---
signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 187.6 on 98 degrees of freedom
Multiple R-squared:  0.05654,    Adjusted R-squared:  0.04692
F-statistic: 5.874 on 1 and 98 DF,  p-value: 0.0172
```

67. Which of the following statements correctly describes the roles of the variables in this regression model?
- A) Response: Age, Explanatory variable: Retinol plasma levels
 - B) Response: Cancer, Explanatory variable: Age
 - C) Response: Retinol plasma levels, Explanatory variable: Age
 - D) Response: Cancer, Explanatory variable: Retinol plasma levels
 - E) Response: Retinol plasma levels, Explanatory variable: Intercept
68. The value of the estimated regression slope $\hat{\beta}_1$ is closest to:
- A) 3.07
 - B) 441.86
 - C) 1.27
 - D) 2.42
 - E) 187.6

69. Which one of the following is the appropriate pair of hypotheses for testing the slope parameter?
- A) $H_0 : \beta_0 = 0, H_A : \beta_0 \neq 0$ D) $H_0 : \hat{\beta}_1 \neq 0, H_A : \hat{\beta}_1 = 0$
 B) $H_0 : \hat{\beta}_1 = 0, H_A : \hat{\beta}_1 \neq 0$ E) $H_0 : \beta_1 = \hat{\beta}_1, H_A : \beta_1 \neq \hat{\beta}_1$
 C) $H_0 : \beta_1 = 0, H_A : \beta_1 \neq 0$
70. The t -value for testing the slope parameter β_1 is $t = 2.424$. The p -value for this test statistic can be calculated by computing $2\Pr(T \geq 2.424)$, where T follows which of the following distributions?
- A) t -distribution with $\nu = 100$ degrees of freedom.
 B) t -distribution with $\nu = 99$ degrees of freedom.
 C) t -distribution with $\nu = 98$ degrees of freedom.
 D) t -distribution with $\nu = 1$ degrees of freedom.
 E) Standard normal distribution.
71. The resultant p -value from the test statistic above is $p = 0.0172$. Select the appropriate interpretation of this result, using $\alpha = 0.05$ as the significance level.
- A) The p -value is greater than our significance level. The data therefore provides some evidence of there being an association between age and retinol plasma levels.
 B) The p -value is less than our significance level. The data therefore provides some evidence that the parameter β_0 is not zero.
 C) The p -value is greater than our significance level. The data therefore provides some evidence of there being *no* association between age and retinol plasma levels.
 D) The p -value is less than our significance level. The data therefore provides some evidence of there being *no* association between age and retinol plasma levels.
 E) The p -value is less than our significance level. The data therefore provides some evidence of there being an association between age and retinol plasma levels.
72. The estimated regression intercept is $\hat{\beta}_0 = 441.86$. Select the most suitable interpretation of this estimate.
- A) $\hat{\beta}_0$ is the mean age of someone with retinol plasma levels of zero.
 B) $\hat{\beta}_0$ is the estimated mean increase in retinol plasma levels for every one year increase in age.
 C) $\hat{\beta}_0$ is the estimated mean retinol plasma levels for a person of the average age across in people in the study.
 D) $\hat{\beta}_0$ is an estimate of the minimum retinol plasma levels that could ever be observed.
 E) $\hat{\beta}_0$ is the estimated mean retinol plasma levels for a new-born baby.

73. Select the most suitable interpretation of what $\hat{\beta}_1$ represents in the model.
- A) $\hat{\beta}_1$ is the true mean increase in retinol plasma levels for every one year increase in age.
 - B) $\hat{\beta}_1$ is the estimated mean increase in retinol plasma levels for every one year increase in age.
 - C) $\hat{\beta}_1$ is the estimated mean decrease in retinol plasma levels for every one year increase in age.
 - D) $\hat{\beta}_1$ is the estimated mean retinol plasma levels for a one-year old.
 - E) $\hat{\beta}_1$ is the estimated mean retinol plasma levels for a person of the average age across in people in the study.
74. Which of the following statements about the fit of the regression model is correct?
- A) Since $R^2 = 5.654\%$ exceeds an $\alpha = 0.05$ significance level, there is evidence of an association between age and retinol plasma levels.
 - B) According to the regression model, 0.05654% of the variation in retinol plasma levels is explained by age.
 - C) According to the regression model, 5.654% of the variation in retinol plasma levels is explained by age.
 - D) Because $\hat{\beta}_1$ is well above 1, there is evidence of an association between age and retinol plasma levels.
 - E) About 5.654% of the variation in retinol plasma levels can be explained, the rest is purely due to randomness across the population.
75. What does the model predict the retinol plasma level of a 50-year old would be? All values are expressed in ng/mL.
- A) 441.86 B) 505.21 C) 444.69 D) 595.36 E) 562.86
76. Let (LB, UB) be the 95% prediction interval for the retinol plasma levels of a 40-year old predicted by the model. Select the appropriate interpretation of this interval.
- A) There is a probability of 0.95 that a randomly selected New Zealand adult has retinol plasma levels between LB and UB ng/mL.
 - B) If we take a 40 year old New Zealand adult, there is a probability of 0.95 that their retinol plasma levels are between LB and UB ng/mL.
 - C) There is a probability of 0.95 that the mean retinol plasma levels across all 40 year old New Zealand adults is between LB and UB ng/mL.
 - D) There is a 5% chance that a randomly selected New Zealand adult has retinol plasma levels above UB ng/mL.
 - E) If a New Zealand adult has retinol plasma levels between LB and UB ng/mL, there is a probability of 0.95 that they are 40 years old.

77. The mean age of the study participants was 50.61 years old. Select the correct statement from the options below about prediction errors.
- A) The prediction error for retinol plasma levels could be reduced to zero for all ages if a large enough sample had been used.
 - B) The sample size has no impact on the size of the prediction error of the retinol plasma levels for all ages.
 - C) The prediction error is smaller when predicting the retinol plasma levels of a 40.61 year old than when predicting the retinol plasma levels of a 50.61 year old.
 - D) The prediction error is larger when predicting the retinol plasma levels of a 40.61 year old than when predicting the retinol plasma levels of a 50.61 year old.
 - E) The prediction error is the same when predicting the retinol plasma levels of a 40.61 year old than when predicting the retinol plasma levels of a 50.61 year old.
78. One 42 year old study participant has retinol plasma levels of 580.7ng/mL. Select the value below which the raw residual for this data point is closest to. All values are expressed in ng/mL.
- A) -9.9 B) 9.9 C) 3.07 D) -3.07 E) 0
79. If the researchers wished to explore whether retinol plasma levels predicted the likelihood of a person getting cancer, which type of regression analysis should they use.
- A) Logistic regression with cancer (yes=1/no=0) as the response variable and retinol plasma levels as the explanatory variable.
 - B) Logistic regression with retinol plasma levels as the response variable and cancer (yes=1/no=0) as the explanatory variable.
 - C) Simple linear regression with cancer (yes=1/no=0) as the response variable and retinol plasma levels as the explanatory variable.
 - D) Simple linear regression with retinol plasma levels as the response variable and cancer (yes=1/no=0) as the explanatory variable.
 - E) ANOVA with cancer (yes=1/no=0) as the response variable and retinol plasma levels as the predictor variable.
-

Information for Questions 80 to 84

Medical insurance companies need to be able to predict their client's medical costs in order to reliably set insurance levies. This is the role of actuaries.

A U.S medical insurance company wants to see if a person's medical costs (**charges**) can be predicted from the variables age (**age**), sex (**sex**), BMI (**bmi**), number of children (**children**), and smoking status (**smoker**).

They run a multiple linear regression analysis in R on the data from $n = 1338$ of their clients. Their results are displayed below.

```
> summary(lm(charges~age+sex+bmi+children+smoker))

Call:
lm(formula = charges ~ age + sex + bmi + children + smoker)

Residuals:
    Min       1Q   Median       3Q      Max
-11837.2  -2916.7   -994.2   1375.3  29565.5

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -12052.46    951.26  -12.670  < 2e-16 ***
age           257.73     11.90   21.651  < 2e-16 ***
sexmale      -128.64    333.36   -0.386  0.699641
bmi           322.36     27.42   11.757  < 2e-16 ***
children      474.41    137.86    3.441  0.000597 ***
smokeryes    23823.39    412.52   57.750  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6070 on 1332 degrees of freedom
Multiple R-squared:  0.7497,    Adjusted R-squared:  0.7488
F-statistic: 798 on 5 and 1332 DF,  p-value: < 2.2e-16
```

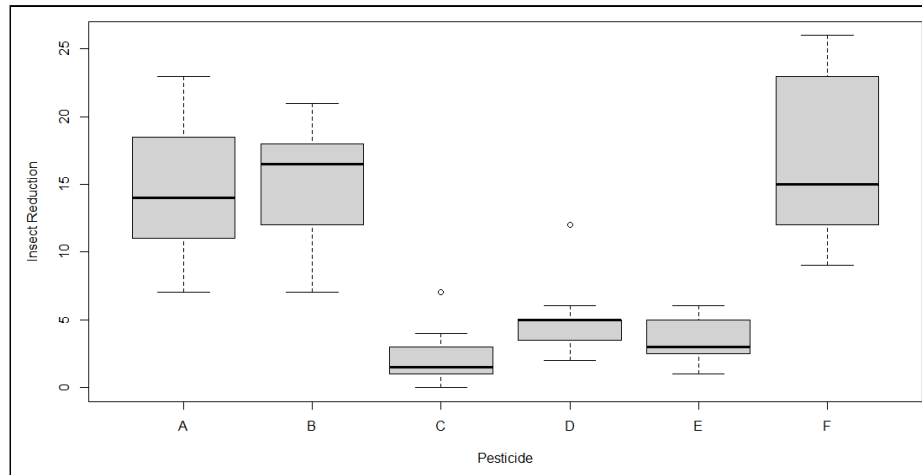
80. Select which variable is the response variable in the model.
- A) **age**
 - B) **sex**
 - C) **bmi**
 - D) **children**
 - E) **charges**
81. In the regression model, a person being a male is coded as **sex**=1 (while **sex**=0 represents that the person is female), while **smoker**=1 indicates the person is a smoker (while **smoker**=0 indicates they are not). Calculate the predicted medical charges for a 50 year old, male, non-smoker, with 3 children, and a BMI of 28kg/m^2 . All values are expressed in dollars.
- A) 34978.1 B) 11154.71 C) 2128.63 D) 9731.48 E) 11283.35

82. Select the most appropriate interpretation of the estimated slope parameter for the BMI variable ($\hat{\beta}_{bmi}$).
- A) $\hat{\beta}_{bmi}$ is the estimated mean medical charge for a client with a **bmi**=1 kg/m².
 - B) $\hat{\beta}_{bmi}$ is the *estimated* mean change in medical charges for every increase of 1 BMI point, assuming **age**, **sex**, **children**, and **smoker** are kept constant.
 - C) $\hat{\beta}_{bmi}$ is the *estimated* mean change in medical charges for every increase of 1 BMI point.
 - D) $\hat{\beta}_{bmi}$ is the *true* mean change in medical charges for every increase of 1 BMI point across this medical insurance company's clients, assuming **age**, **sex**, **children**, and **smoker** are kept constant.
 - E) $\hat{\beta}_{bmi}$ is the *true* mean change in medical charges for every increase of 1 BMI point across this medical insurance company's clients.
83. The 95% confidence interval for the slope coefficient of age (β_{age}) is calculated using:
- A) $257.73 \pm t_{0.975,1332} \times 11.9$
 - B) $257.73 \pm t_{0.975,1337} \times \frac{6070}{\sqrt{1338}}$
 - C) $257.73 \pm t_{0.975,1337} \times 11.9$
 - D) $257.73 \pm t_{0.975,1332} \times \frac{6070}{\sqrt{1338}}$
 - E) $257.73 \pm t_{0.975,1336} \times \frac{6070}{\sqrt{1338}}$
84. The coefficient of determination in the model is $R^2 = 0.7497$. Select the appropriate interpretation of this result from the options below.
- A) The model explains about 74.97% of the variation in the **age**, **sex**, **bmi**, **children**, and **smoker** variables.
 - B) The variable **age** explains about 74.97% of the variation in the medical charges.
 - C) The variables **age**, **sex**, **bmi**, **children**, and **smoker** explain about 74.97% of the variation in medical charges.
 - D) Around 74.97% of the variation in medical charges can be predicted. The other 25.03% of the variation is not explainable and is purely caused by random variation.
 - E) The relationship between medical charges and the **age**, **sex**, **bmi**, **children**, and **smoker** variables is about 74.97% linear.
-

Information for Questions 85 to 92

A study is conducted to test the effectiveness of 6 different pesticides (labelled 'A' through 'F') at reducing insect numbers on cropped farmland. 72 small segments of various crop fields were portioned off. Each of the 72 segments were allocated one pesticide, with each of the six pesticides being allocated to 12 fields. For each segment, the number of insects on the segment was recorded before the pesticide was applied, and after one month of the pesticide being applied daily.

A box plot for the reduction in the number of insects between before and after the pesticide was applied for each of the pesticides is displayed below.



The data were analysed in R, producing the output below.

```
> anova(lm(reduction~pesticide))
Analysis of Variance Table

Response: reduction
      Df Sum Sq Mean Sq F value    Pr(>F)
pesticide  5 2668.8   533.77  34.702 < 2.2e-16 ***
Residuals 66 1015.2    15.38
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

85. The ANOVA model used here can be written as

$$Y_{ij} = \mu_i + e_{ij}$$

where Y_{ij} represents the random variable for the reduction in the number of insects in the j^{th} field segment that was applied the i^{th} pesticide. Select what μ_i and e_{ij} represent in the above equation from the options below.

- A) μ_i is the mean number of insects on segments that were applied pesticide i before the pesticide was applied in our sample; e_{ij} is the mean reduction in the number of insects in field segments that were applied pesticide i in our sample.
 - B) μ_i is the mean reduction in the number of insects in field segments that were applied pesticide i in our sample; e_{ij} is the 'residual', or difference between the reduction in the number of insects observed in the j^{th} field segment that was applied pesticide i , and μ_i .
 - C) μ_i is the population-level mean reduction in the number of insects on field segments that are applied pesticide i ; e_{ij} is the 'residual', or difference between the reduction in the number of insects observed in the j^{th} field segment that was applied pesticide i , and μ_i .
 - D) μ_i is the mean reduction in the number of insects in field segments that were applied pesticide i in our sample; e_{ij} is the standard deviation of the j^{th} observation in group i .
 - E) μ_i is the population-level mean reduction in number of insects on field segments that are applied pesticide i ; e_{ij} is the standard deviation of the j^{th} observation in group i .
86. Which of the following statements regarding the e_{ij} term in the ANOVA model is correct?
- A) The e_{ij} values for each group i are assumed to normally distributed with a zero mean and constant variance.
 - B) The e_{ij} values for each group i are assumed to follow a t -distribution.
 - C) The e_{ij} values are only approximately normal if the sample sizes for each group of observations are the same.
 - D) The e_{ij} terms do not need to follow any particular distribution, but the overall sample size must be large enough such that the e_{ij} terms follow a t -distribution.
 - E) The e_{ij} terms do not need to follow any particular distribution, but the sample size is large enough that the e_{ij} terms follows a normal distribution.
 - F) The e_{ij} terms do not need to follow any particular distribution, but the sample size is large enough that the sampling distribution for the mean of the e_{ij} terms follows a t -distribution.

87. Select the hypothesis that is being tested by the F -statistic in the ANOVA table above.
- A) $H_0: \mu_A = \mu_B = \mu_C = \mu_D = \mu_E = \mu_F$; H_A : All the means are different to each other.
 - B) H_0 : All the means are different to each other; H_A : All the means are the same.
 - C) $H_0: \mu_A = \mu_B = \mu_C = \mu_D = \mu_E = \mu_F$; $H_A: \mu_A, \mu_B, \mu_C, \mu_D, \mu_E, \mu_F$ are not all equal.
 - D) $H_0: \hat{\mu}_A = \hat{\mu}_B = \hat{\mu}_C = \hat{\mu}_D = \hat{\mu}_E = \hat{\mu}_F$; $H_A: \hat{\mu}_A, \hat{\mu}_B, \hat{\mu}_C, \hat{\mu}_D, \hat{\mu}_E, \hat{\mu}_F$ are not all equal.
 - E) H_0 : The sample means are all equal to the population means; H_A : The sample means are not all equal to the population means.
88. The F -statistic in the ANOVA analysis is based on various sums of squares (SS). Select the correct statement below regarding the sums of squares in this analysis.
- A) The total SS is 2668.8; the residual SS is 1015.2.
 - B) The total SS is 3684.0; the residual SS is 1015.2.
 - C) The group SS is 1015.2. The residual SS is 2668.8.
 - D) The group SS is 3684.0; the residual SS is 1015.2.
 - E) The total SS is 3684.0; the group SS is 1015.2.
89. Select the circumstance under which we would be *most* likely to reject the null hypothesis in a one-way ANOVA analysis.
- A) When the variability between the group means is small, and the variability within the study groups is small.
 - B) When the variability between the group means is small, and the variability within the groups is large.
 - C) When the variability between the group means is large, and the variability within the groups is large.
 - D) When the variability between the group means is large, and the variability within the groups is small.
 - E) When the overall variability across all of the group observations is large.
90. Letting **f** represent the F -statistic calculated in this ANOVA, select the appropriate R command to calculate the p -value from the ANOVA table.
- A) **pf(f,71,lower.tail=FALSE)**
 - B) **1-pf(f,71,lower.tail=TRUE)**
 - C) **pf(f,5,66,lower.tail=FALSE)**
 - D) **1-pf(f,5,66,lower.tail=FALSE)**
 - E) **2*(pf(f,5,66,lower.tail=FALSE))**

91. Using $\alpha = 0.01$ as the significance level, the most appropriate conclusion to draw from the p -value in the ANOVA table above is:
- A) Since our p -value is greater than $\alpha = 0.01$, we do not have statistically significant evidence to reject H_0 , and we do not have statistically significant evidence that the mean reduction in the number of insects differs between some of the pesticide types.
 - B) Since our p -value is less than $\alpha = 0.01$, we have statistically significant evidence at the 0.01 significance level to reject H_0 . Since the observed sample means (going from smallest to largest) were for pesticide C, E, D, A, F, and B, we have evidence that B is the most effective pesticide at reducing insect numbers, followed by F, A, D, E, then C.
 - C) Since our p -value is less than $\alpha = 0.01$, we have statistically significant evidence at the 0.01 significance level to reject H_0 and conclude that the mean reduction in number of insects differs between some of the pesticide types.
 - D) Since our p -value is less than $\alpha = 0.01$, we do not have statistically significant evidence at the 0.01 significance level to reject H_0 , and we do not have statistically significant evidence that the mean reduction in number of insects differs between pesticide types.
 - E) Since our p -value is less than $\alpha = 0.01$, we have statistically significant evidence at the 0.01 significance level to accept H_0 and conclude that the mean reduction in number of insects is the same between the different pesticide types.
92. It turns out that the field segments were in four different countries, with three field segments from each country being used to test each pesticide. Suppose that the insects from the four different countries have very different resiliencies to pesticides. The following statements concern the possibility of including "country" as an additional variable in the analysis. Which of them is correct?
- A) Including country as a blocking variable in the analysis is likely to soak up some of the total variation across the entire sample, helping to detect differences that exist between the effectiveness of the pesticides.
 - B) Including country as a blocking variable in the analysis is likely to soak up some of the residual variation, helping to detect differences that exist between the effectiveness of the pesticides.
 - C) Including country as a blocking variable in the analysis is likely to soak up some of the residual variation, making it harder to detect differences that exist between the effectiveness of the pesticides.
 - D) Including country as a blocking variable in the analysis is likely to soak up some of the variation between groups, helping to detect differences that exist between the effectiveness of the pesticides.
 - E) Including country as a blocking variable in the analysis is likely to soak up some of the variation between groups, making it harder to detect differences that exist between the effectiveness of the pesticides.

Summary of Formulae

Sample mean and variance

$$\text{Mean: } \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{Variance: } s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Probability Rules

$$\Pr(A \text{ or } B) = \Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B) \quad \Pr(A \text{ and } B) = \Pr(A \cap B) = \Pr(A) \Pr(B|A) \\ = \Pr(B) \Pr(A|B)$$

Random Variables If X and Y are independent random variables, then $W = aX + bY + c$ has:

$$\text{Mean: } \mu_W = a\mu_X + b\mu_Y + c \quad \text{Variance: } \sigma_W^2 = a^2\sigma_X^2 + b^2\sigma_Y^2$$

Discrete Distributions

$$\text{Mean: } \mu_X = \sum_{i=1}^k x_i \Pr(X = x_i) \quad \text{Variance: } \sigma_X^2 = \sum_{i=1}^k (x_i - \mu_X)^2 \Pr(X = x_i)$$

Binomial Distribution

$$\mu_X = n\pi \quad \sigma_X^2 = n\pi(1-\pi) \quad \Pr(X = x) = \binom{n}{x} \pi^x (1-\pi)^{n-x} \quad \binom{n}{x} = \frac{n!}{x!(n-x)!}$$

If $n\pi \pm 3\sqrt{n\pi(1-\pi)}$ are between 0 and n , then X is approximately normally distributed with mean μ_X and variance σ_X^2 .

Normal Distribution A standard normal random variable, Z , has $\mu_Z = 0$ and $\sigma_Z^2 = 1$. To transform a normal random variable X into a standard normal (and vice versa):

$$Z = \frac{X - \mu_X}{\sigma_X} \quad X = Z\sigma_X + \mu_X$$

Distributions of Statistics

- The mean $\bar{X}$ of a random sample of size n has mean $\mu_{\bar{X}} = \mu_X$ and standard error $\sigma_{\bar{X}} = \frac{\sigma_X}{\sqrt{n}}$.
- The sample proportion P computed from a binomial distribution with parameters n and π has a mean of $\mu_P = \pi$ and standard error $\sigma_P = \sqrt{\frac{\pi(1-\pi)}{n}}$. If $n\pi \pm 3\sqrt{n\pi(1-\pi)}$ are between 0 and n , then P will be approximately normally distributed.
- The distribution of the difference between two sample means $\bar{X}_1 - \bar{X}_2$ has a mean of $\mu_{\bar{X}_1 - \bar{X}_2} = \mu_1 - \mu_2$ and a standard error of $\sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$.

Contingency Tables

Factor 1	Factor 2		Total
	Level 1	Level 2	
Level 1	a	b	$r_1 = a + b$
Level 2	c	d	$r_2 = c + d$
	$c_1 = a + c$	$c_2 = b + d$	$n = a + b + c + d$

$$\chi^2 = \sum_{i=1}^R \sum_{j=1}^C \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

- R and C are the number of rows and columns respectively
- $e_{ij} = \frac{r_i c_j}{n}$, where r_i is the i th row total and c_j is the j th column total
- o_{ij} is the observed value in row i column j

With Factor 2 as “Outcome” variable:

$$\text{Odds ratio: } OR = (a/b)/(c/d) = ad/bc$$

$$\text{Relative risk: } RR = (a/r_1) / (c/r_2)$$

$$\text{Risk difference: } RD = a/r_1 - c/r_2$$

Confidence Intervals and Hypothesis Tests

All of the $100(1 - \alpha)\%$ confidence intervals calculated in this course are of the form:

Estimate $\pm$ multiplier $\times$ standard error

In the table $\bar{x}$, p etc are the values calculated from the samples.

	Estimate	df (ν)	Multiplier	Standard Error
Population mean				
• Random sample, σ_X known	$\bar{x}$	NA	$z_{(1-\alpha/2)}$	$\frac{\sigma_X}{\sqrt{n}}$
• Normal population, σ_X unknown	$\bar{x}$	$n - 1$	$t_{(1-\alpha/2, \nu)}$	$\frac{s}{\sqrt{n}}$
• Large random sample ($n \geq 20$), σ_X unknown	$\bar{x}$	$n - 1$	$t_{(1-\alpha/2, \nu)}$	$\frac{s}{\sqrt{n}}$
Difference between population means				
• Large random samples (both ≥ 20)	$\bar{x}_1 - \bar{x}_2$	Will be provided	$t_{(1-\alpha/2, \nu)}$	$\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
• Paired difference in random samples from a normal population	$\bar{d}$	$n - 1$	$t_{(1-\alpha/2, \nu)}$	$\frac{s_d}{\sqrt{n}}$
Population proportions				
• Population proportion	p	NA	$z_{(1-\alpha/2)}$	$\sqrt{\frac{p(1-p)}{n}}$
• Difference between 2 population proportions	$p_1 - p_2$	NA	$z_{(1-\alpha/2)}$	$\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$
• Difference between 2 population proportions (hypothesis test)	$p_1 - p_2$	NA	$z_{(1-\alpha/2)}$	$\sqrt{\frac{p^*(1-p^*)}{n_1} + \frac{p^*(1-p^*)}{n_2}}$
(Use $p^* = \frac{x_1 + x_2}{n_1 + n_2}$ for hypothesis test)				
Odds ratio, relative risk, risk difference (see contingency table on previous page for a, b, c and d)				
• Log odds ratio	$\ln(OR)$	NA	$z_{(1-\alpha/2)}$	$\sqrt{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}}$
• Log relative risk	$\ln(RR)$	NA	$z_{(1-\alpha/2)}$	$\sqrt{\frac{1}{a} - \frac{1}{r_1} + \frac{1}{c} - \frac{1}{r_2}}$
• Risk difference – as for the difference between two population proportions with $p_1 = a/r_1$ and $p_2 = c/r_2$				
After ANOVA and Regression				
• Estimate, multiplier and standard error determined from output				

Regression

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x \text{ where } \hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} \text{ and } \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

$$\text{Standard error of the slope is } s_{\hat{\beta}_1} = \frac{s_e}{\sqrt{\sum(x_i - \bar{x})^2}} \text{ where } s_e = \sqrt{\frac{\sum(y_i - \hat{y}_i)^2}{n - 2}} = \sqrt{\frac{\text{RSS}}{n - 2}}$$

$$\text{Standard error of a prediction at } X = x_0 \text{ is } PE(\hat{y}_0) = s_e \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum(x_i - \bar{x})^2}}$$

ANOVA

$$y_{ij} = \mu_i + e_{ij} \quad \hat{\mu}_i = \bar{y}_i.$$

Post ANOVA analysis

$$s_e = \sqrt{RMS}$$

$$\underbrace{\text{TSS}}_{\text{Total sum of squares}} = \underbrace{\text{GSS}}_{\text{Group sum of squares}} + \underbrace{\text{RSS}}_{\text{Residual sum of squares}}$$